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Key Points:

- Significant geomagnetically induced currents (GIC) intensities were observed and estimated in the Mexican power grid during the geomagnetic storm on 10 May 2024
- The observations suggest that space weather can pose a potential threat to low latitude power grids during extreme geomagnetic storms
- These GIC observations are unique in Latin America, recorded during the most significant geomagnetic storm in the last 20 years

Supporting Information:

Supporting Information may be found in the online version of this article.

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The Impact of Geomagnetically Induced Currents (GIC) on the Mexican Power Grid: Numerical Modeling and Observations From the 10 May 2024, Geomagnetic Storm

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Abstract This study examines the impact of the 10 May 2024, geomagnetic storm on the Mexican power grid, utilizing geomagnetically induced currents (GIC), measurements and regional magnetic field data recorded by the Laboratorio Nacional de Clima Espacial. Significant GIC were observed at three different locations within the grid. The observations were complemented with estimates for the Mexican power grid provided by a numerical model developed in late 2022. Our findings suggest that the GIC can pose a potential threat to low-latitude power grids during extreme geomagnetic disturbances. Furthermore, the model demonstrates its potential to forecast the grid response during these events, providing critical insight into the behavior of the electrical grid during extreme space weather events.

Plain Language Summary The geomagnetic storm occurred on 10 May 2024, was the most intense space weather event since the Halloween storm of October 2003. In this case, we present observations and estimates of GIC in the Mexican power grid during the event. Significant GIC intensities were measured at some substations of the Mexican power grid during the most active times. Our findings suggest that space weather effects like GIC can pose a potential threat to the integrity of power grids during extreme geomagnetic disturbances, including those across the Americas.

1. Introduction

On 8 May 2024, the active region AR3664 emitted a series of X-class solar flares and seven halo coronal mass ejections, which impacted Earth's magnetosphere on 10-11 May. These solar events triggered the strongest geomagnetic storm since the Halloween storm of 2003 (Balch et al., 2004). The enhanced magnetospheric convection and energy influx over the upper atmosphere led to strong thermospheric heating, changes in the ionospheric O/N_2 ratio and dynamics (Bojilova et al., 2024; Kwak et al., 2024). Additionally, disruptions in farming precision navigational systems, power grids and flight and satellite operations were reported (Evans et al., 2024; O'Callaghan & Billings, 2024; Parker & Linares, 2024). The strong particle precipitation generated auroras that were observed even at low latitudes in both hemispheres during the night of 10-11 May. Particularly, in Mexico, red auroras were visible between 8 p.m. and 4 a.m. local time, with reports from about half of its territory (Gonzalez-Esparza et al., 2024).

In recent years, the Laboratorio Nacional de Clima Espacial (LANCE), at the Universidad Nacional Autónoma de México (UNAM), has been developing instrument networks to monitor the regional effects of space weather in Mexico (Gonzalez-Esparza et al., 2016). These networks include geomagnetic stations (GMS) and GIC, sensors at key 400 kV power grid substations, the latter in collaboration with the national power utility, the Comisión Federal de Electricidad (CFE). The GIC are a direct effect of geomagnetic disturbances on technological systems at Earth's surface and might pose a threat to power grids, even those located at low-to-mid latitudes (Carter et al., 2015; Gaunt, 2014).

Though GIC effects on high-latitude power grids has been reported since the mid of the twentieth century (McNish, 1940), their effects on low-latitude ones still remain difficult to assess. The GIC research in low-latitude

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technological systems is still scarce, having little observational evidence reported in the last 25 years (Barbosa et al., 2015; Bernhardt et al., 2008; Trivedi et al., 2007).

2. Modeling GIC in the Mexican Power Grid

The GIC arise from geomagnetic disturbances produced by the magneto-ionospheric currents. According to Faraday's law, a time-varying geomagnetic field induces a geoelectric field $\mathbf{E}$ at the Earth's surface. In absence of free charges inside the integration volume, $\mathbf{E} = -\partial\mathbf{A}/\partial t$, being $\mathbf{A}$, the magnetic vector potential. Conversely, in the frequency domain, it can be calculated from the geomagnetic field $\mathbf{B}$ as:

$$E(\omega)_{x,y} = \pm \frac{1}{\mu_0} Z(\omega) B(\omega)_{y,x} \quad (1)$$

where Z is the ground impedance tensor, μ_0 the magnetic permeability and (x,y) subscripts representing northward and eastward components, respectively.

Then, the geovoltage induced along the power lines (i.e., between network nodes), is calculated by integrating along them as:

$$V_{ij} = \int_i^j \mathbf{E} \cdot d\mathbf{s} \quad (2)$$

with $d\mathbf{s}$ the line element.

Due to the quasi-DC nature of the GIC, the power grid is considered as a series-parallel array of resistances and power sources. The power sources represent the geovoltage induced, as in Equation 2 whilst the resistance represent the conductor resistances (Figure S1 in Supporting Information S1). Then, we can apply the Kirchhoff's rules to solve for the currents at each branch. In practice, we use the Nodal Admittance Method to find the currents at the grounding points of the grid (Lehtinen & Pirjola, 1985). Interested readers can obtain further details about the GIC calculation in Supporting Information S1 and in Boteler and Pirjola (2017).

Caraballo et al. (2020) presented a numerical model to estimate GIC for 115 substations of the 400 kV Mexican power grid. As a first approximation, it assumed uniform ground conductivity across the Mexican territory and a spatially uniform geomagnetic disturbance. The initial results suggested that the Mexican power grid could experience GIC between 20 and 75 A during severe or extreme geomagnetic storms. These initial results helped to identify the most vulnerable substations, guiding future installations of in-situ GIC monitoring equipment.

Subsequently, Caraballo et al. (2023) improved the GIC numerical model, by including different voltage levels (400/230 kV). Currently, the model calculates GIC for a network of 246 substations, comprising 99% of the 400 kV grid and 45% of the 230 kV grid. This model uses elementary spherical current systems (Amm, 1997), to interpolate the geomagnetic disturbance (ΔB) at locations far away of the GMS. In addition, employed a 1D piecewise model to consider lateral variations in ground conductivity. In this case, we integrate geomagnetic field, ground conductivity and power grid data to provide GIC estimates for a significant portion of the 400/230 kV grid. A brief description of the numerical model is given in Supporting Information S1.

The study compared observed and calculated GIC at a coastal 400 kV substation between August and November 2021. Five geomagnetic storms occurred in this period (ranging from G1 to G2). During the strongest event, the observed GIC exceeded 10 A. Finally, the paper emphasized the need for additional regional geomagnetic field measurements, as well as more accurate data describing the ground conductivity and the power grid parameters to improve the results. However, the model yield reasonable estimates despite all these shortcomings. The latter can be ascribed to the fact that sometimes GIC intensities are more dependent on the grid's characteristics (i.e., resistances, geographic orientation, topology, etc.), rather than on the ground conductivity, which can be significant over short distances only. Hence, uncertainties in grid's characteristics can imply up-to a 100% change in the final results whilst ground conductivity ones barely amount by 40% change in most cases (Beggan, 2015; Caraballo, 2016). Fortunately, power grid parameters are a relatively accessible part of the input. Anyway, this is a milestone for GIC research in Latin America.

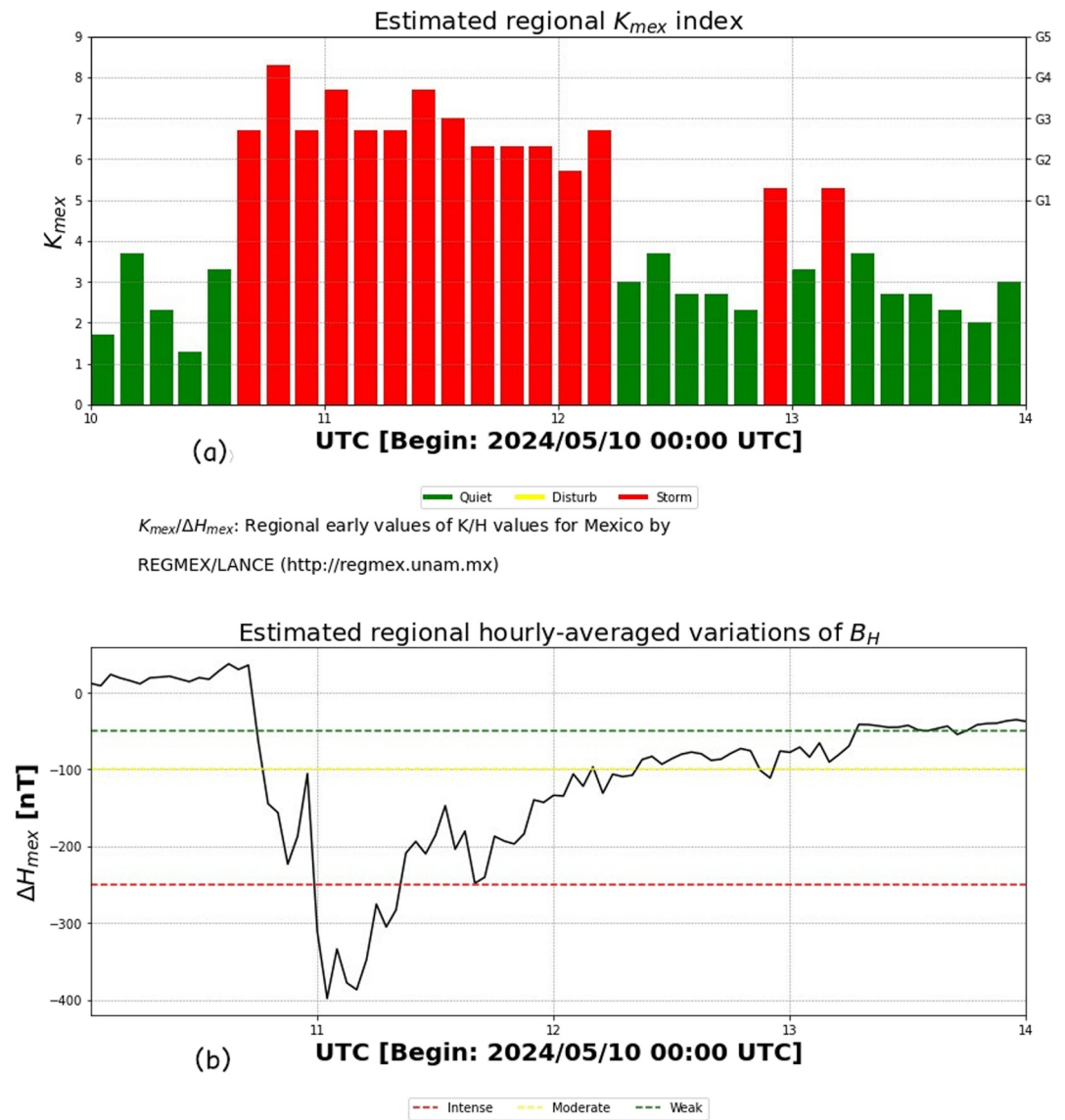


Figure 1. (a) Regional K_{mex} index covering the full storm period and (b) the estimated variations in the horizontal component of the geomagnetic field in Mexico, ΔH .

2.1. The Magnetic Stations Network

The LANCE's GMS Network of Mexico project (REGMEX), aims to develop an array of GMS across Mexico to monitor the regional effects of space weather on the country's infrastructure (Corona-Romero et al., 2024). Two GMS stations recorded the geomagnetic storm on 10 May 2024: one located at Coeneo (COE), Michoacán (27.71° N, 31.08° W), and another at Iturbide (ITU), Nuevo León (35.69° N, 29.87° W). Additionally, LANCE collaborates with the Magnetic Service of the Geophysics Institute and the Center of Geosciences at UNAM, providing data from the Magnetic Observatory of Teoloyucan (TEO), State of Mexico (27.84° N, 28.41° W), and the Juriquilla magnetotellurics (MT) station (28.7° N, 29.91° W), respectively.

Figure 1 shows the evolution of the regional K_{mex} geomagnetic index and the variation in the horizontal component of the geomagnetic field in Mexico, ΔH , during the Mother's Day geomagnetic storm. The figure shows that ΔH dropped to -400 nT at the storm main phase, with $K_{mex} \gtrsim 8$ after the storm onset. Moreover, we can see that, K_{mex} and ΔH remained $\gtrsim 6$ and ≤ -100 nT by ≈ 36 hrs, respectively. This corresponds to $\sim 38\%$ of the entire storm time.

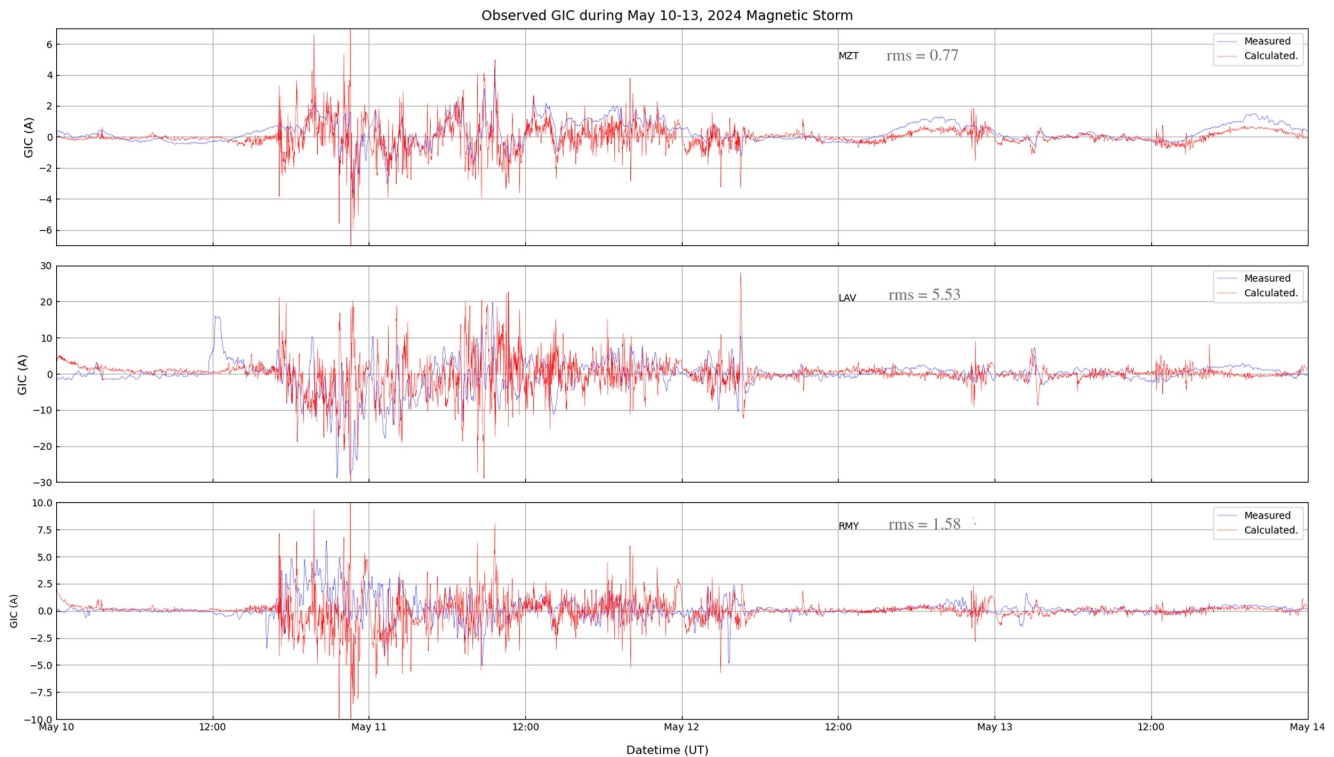


Figure 2. Observed (blue) and calculated geomagnetically induced currents (red) at three substations of the Mexican 400/230 kV power grid.

3. GIC Observations and Numerical Results

Currently, there are three GIC sensors installed at the neutrals of high-voltage transformer banks at the following 400 kV substations: Mazatlán, Sinaloa (MZT, 23.19° N, 106.35° W), Laguna Verde, Veracruz (LAV, 19.72° N, 96.41° W), and Riviera Maya, Quintana Roo (RMY, 20.83° N, 86.93° W). The sensors consist of a Hall detector, an analog-to-digital converter and a data logger to digitize the signal and deliver the data stream to CFE. A description of the GIC sensors is provided in Section 2 in Supporting Information S1.

This paper presents observed GIC from these three substations and estimates for 246 substations of the 400/230 kV grid. Some sensors have been operational for nearly 5 years, providing 1 min GIC data, which is periodically processed at LANCE.

We used data from REGMEX stations, as well as INTERMAGNET observatories of Tucson (TUC, 32.17° N, 110.73° W) and Stennis Space Center (BSL, 30.35° N, 89.64° W), to calculate GIC in the Mexican power grid. The augmented geomagnetic coverage improved the interpolation of the geomagnetic disturbance ΔB , along the Mexican territory, a feature not present in previous studies. In this sense, the regional data was valuable to catch the local geomagnetic variability.

Figure 2 shows observed and calculated GIC at MZT, LAV, and RMY substations. The root-mean-square (RMS) errors of the estimates are also shown. We can see that the agreement between observed and calculated values do not exceed 2 A in the RMY and MZT case. This is within the error interval of the calculated data. The greater RMS error in the LAV case can be ascribed by the initial jump around 12:00 UT, which corresponds to an instrumental glitch, unrelated to actual geomagnetic activity. Conversely, the LAV and RMY records show a noticeable impulse at the storm onset (around 17:30 UT on 10 May), which is larger in the LAV case. Regarding the observed GIC at LAV substation, it peaked ~ 30 A during the storm main phase and often exceeding by $>50\%$ the estimates of the numerical model.

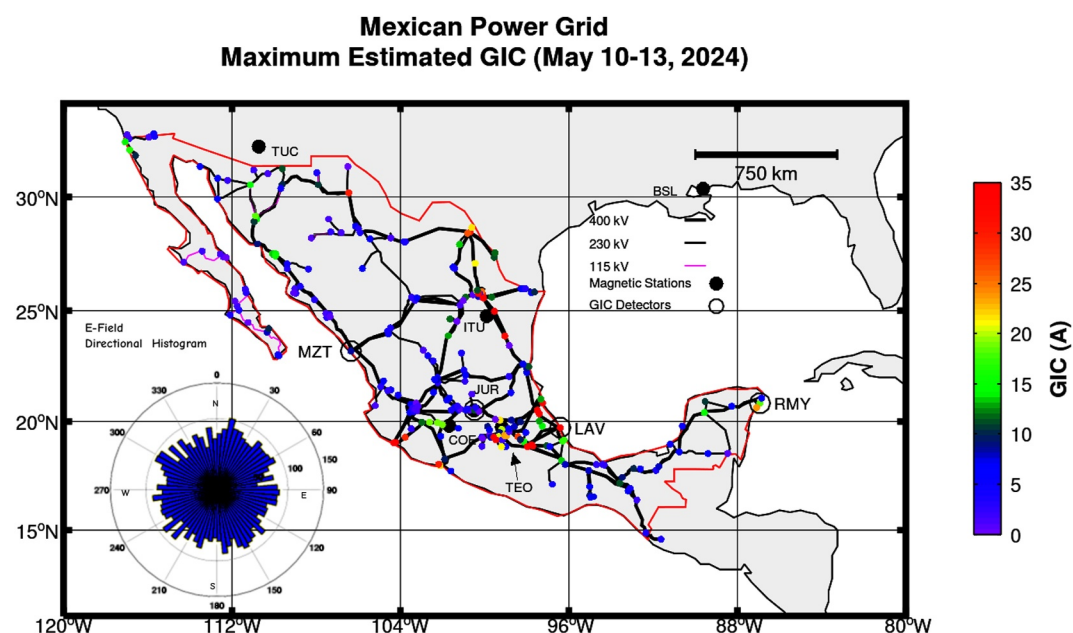


Figure 3. Maximum estimated geomagnetically induced currents (GIC) in all the substations considered in the 400/230 kV power grid. Magnetic observatories/stations are marked as black dots, and GIC sensor sites are marked with circles. 400/230/115 kV power lines are marked in different styles. The wind rose show the directional histogram of the induced E-field along the central Mexican territory during the event.

4. Discussion

According to Figure 2, the grid response to the geomagnetic storm reached at least twice the maximum GIC previously reported for low latitudes (Barbosa et al., 2015; Bernhardt et al., 2008; Caraballo et al., 2023). Unfortunately, for this event, we were unable to find similar low-latitude GIC data to compare our observations. This makes these observations more interesting. However, CFE reported minimal impact on grid operations in this case, presumably due to the brief duration of the disturbances. This agrees with some reports published soon after the storm (O'Callaghan & Billings, 2024) and contrast strongly with the failures caused by the Halloween storms on the South African power grid in 2003–2004 (Gaunt & Coetzee, 2007).

The differences between observed GIC and model predictions can be attributed to the aforementioned uncertainties in some input parameters like local ground conductivity, unaccounted local geomagnetic variability, instrumental issues, etc. Since GIC sensors are placed at outdoors and constantly exposed to intense electromagnetic noise, they are not 100% immune to external influences.

Input parameters, such as substation grounding resistances and local ground conductivity, remain poorly characterized across much of Mexico. Future MT surveys could provide more accurate data. Nevertheless, the numerical model yields acceptable estimates of GIC levels at the considered substations. This is an iterative procedure, as more accurate data will be available we will get better estimates.

Conversely, we observe that GIC peak intensities at LAV are larger than the other substations, despite all of them are located in coastal areas. This can be due to a particular combination of local features like ground properties and network topology.

Figure 3 illustrates the maximum GIC intensities calculated for 246 substations of the 400/230 kV power grid. Reddish points, indicating higher GIC peaks, are concentrated at the grid's edges and some coastal substations. Moreover, the most affected substations are those working at 400 kV, connecting long branches mainly in the N-S direction, this makes sense since there is a proportional relation between operational voltage, branch lengths and GIC intensity (Kappenman, 2007). Additionally, the model suggests significant GIC activity in central Mexico, a novel feature not seen in previous studies. Though, the histogram of the induced E-field directions shows a quite homogeneous distribution, some maxima align closely parallel to the main axes of the grid. Moreover, Figure 3 shows a slight similitude with Figure 6b in Caraballo et al. (2020).

Finally, these observations can serve as an early warning about the threats that extreme magnetic storms might pose over low-latitude power grids. Factors, like interconnections between national grids and their natural trend to expand, may amplify these effects in unexpected ways.

5. Conclusions

The GIC registered in the Mexican power grid during the Mother's Day geomagnetic storm allow assessing the vulnerability of strategic facilities and infrastructure located at low-to-mid latitudes. In Mexico, GIC intensities of ~ 30 A were observed at some 400 kV substation, with effects lasting up to 2 hours. Although these effects are not sufficient to cause catastrophic damage, the cumulative effects of this type of event are difficult to determine. When comparing observational versus calculated GIC, we found quantitative and qualitative consistencies. Such consistencies help us to validate and improve the numerical model and to achieve a better comprehension of the Mexican power grid response to severe geomagnetic storms. This study serves as an early warning of potential severe space weather events during the remainder of the current solar cycle. Finally, this is the only GIC research project reporting space weather effects in a Latin American power grid since 2019.

Data Availability Statement

The authors comply with AGU's data policy regarding FAIR Data standards. The Readers can access the GNU Octave and Python scripts used in this paper at the OSF general repository (Caraballo, 2024), folder "GIC_assess: GRL Article 2024/scripts." GNU Octave, it is free software under the terms of the GNU General Public License. The results presented in this paper are from processed data from the INTERMAGNET magnetic observatories of Tucson and Stennis Space Center. This data can be accessed at www.intermagnet.org. Additional magnetic data from REGMEX stations was obtained from (<http://regmex.unam.mx>). The magnetic data used in this study is located in folder "GIC_assess:GRL Article 2024/datasets."

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References

- Amm, O. (1997). Ionospheric elementary current systems in spherical coordinates and their application. *Journal of Geomagnetism and Geoelectricity*, 49(7), 947–955. <https://doi.org/10.5636/jgg.49.947>
- Balch, C., Biesecker, D., Combs, L., Crown, M., Doggett, K., & Kunches, J. D. Z. (2004). *NOAA technical memorandum OAR SEC-88 HALLOWEEN space weather storms of 2003 (tech. Rep.)*. UNITED STATES DEPARTMENT OF COMMERCE, NATIONAL OCEANIC AND ATMOSPHERIC ADMINISTRATION. Retrieved from http://www.swpc.noaa.gov/Services/HalloweenStorms_assessment.pdf
- Barbosa, C., Alves, L., Caraballo, R., Hartmann, G., Papa, A., & Pirjola, R. (2015). Analysis of geomagnetically induced currents at a low-latitude region over the solar cycles 23 and 24: Comparison between measurements and calculations. *Journal Space Weather Space Climate*, 5, A35. <https://doi.org/10.1051/swsc/2015036>
- Beggan, C. (2015). Sensitivity of geomagnetically induced currents to varying auroral electrojet and conductivity models. *Earth Planets and Space*, 67(1), 24. <https://doi.org/10.1186/s40623-014-0168-9>
- Bernhardi, E. H., Cilliers, P. J., & Gaunt, C. T. (2008). Improvement in the modelling of geomagnetically induced currents in Southern Africa. *South African Journal of Science*, 104, 265–272.
- Bojilova, R., Mukhtarov, P., & Pancheva, D. (2024). Global ionospheric response during extreme geomagnetic storm in may 2024. *Remote Sensing*, 16(21), 4046. <https://doi.org/10.3390/rs16214046>
- Boteler, D. H., & Pirjola, R. J. (2017). Modeling geomagnetically induced currents. *Space Weather*, 15(1), 258–276. Retrieved from <https://doi.org/10.1002/2016SW001499>
- Caraballo, R. (2016). Geomagnetically induced currents in Uruguay: Sensitivity to modelling parameters. *Advances in Space Research*, 58(10), 2067–2075. Retrieved from <https://doi.org/10.1016/j.asr.2016.03.006>
- Caraballo, R. (2024). Gic assessment on the mexican power grid [dataset]. *Open Science Framework*. Retrieved from <https://doi.org/10.17605/OSF.IO/AV9HT>
- Caraballo, R., González-Esparza, J. A., Pacheco, C. R., & Corona-Romero, P. (2023). Improved model for gic calculation in the mexican power grid. *Space Weather*, 21(10). Retrieved from <https://doi.org/10.1029/2022SW003202>
- Caraballo, R., González-Esparza, J. A., Sergeeva, M., & Pacheco, C. R. (2020). First GIC estimates for the Mexican power grid. *Space Weather*, 18(2). Retrieved from <https://doi.org/10.1029/2019SW002260>
- Carter, B. A., Yizengaw, E., Pradipta, R., Halford, A. J., Norman, R., & Zhang, K. (2015). Interplanetary shocks and the resulting geomagnetically induced currents at the equator. *Geophysical Research Letters*, 42(16), 6554–6559. Retrieved from <https://doi.org/10.1002/2015GL065060>
- Corona-Romero, P., Gonzalez-Esparza, J., Villanueva-Hernandez, P., Andrade-Mascote, E., Castellanos-Velazco, C., Espinosa-Jimenez, A., & Sergeeva, M. (2024). The geomagnetic stations network of Mexico (regmex): The initial steps towards a real-time monitoring of geomagnetic activity in Mexico. *Journal of Atmospheric and Solar-Terrestrial Physics*, 256, 106204. Retrieved from <https://doi.org/10.1016/j.jastp.2024.106204>
- Evans, J. S., Correia, J., Lumpe, J. D., Eastes, R. W., Gan, Q., Laskar, F. I., et al. (2024). Gold observations of the thermospheric response to the 10?12 may 2024 gannon superstorm. *Geophysical Research Letters*, 51(16). <https://doi.org/10.1029/2024GL110506>
- Gaunt, C. T. (2014). Reducing uncertainty - Responses for electricity utilities to severe solar storms. *Journal Space Weather Space Climate*, 4(27), A1. <https://doi.org/10.1051/swsc/2013058>
- Gaunt, C. T., & Coetzee, G. (2007). Transformer failures in regions incorrectly considered to have low GIC-risk. In *2007 IEEE Lausanne Power Tech* (pp. 807–812). <https://doi.org/10.1109/PCT.2007.4538419>

- Gonzalez-Esparza, J. A., De la Luz, V., Corona-Romero, P., Mejia-Ambriz, J. C., Gonzalez, L. X., Sergeeva, M. A., et al. (2016). Mexican space weather Service (SCIESMEX). *Space Weather*, 15(1), 3–11. Retrieved from <https://doi.org/10.1002/2016SW001496>
- Gonzalez-Esparza, J. A., Sanchez-Garcia, E., Sergeeva, M., Corona-Romero, P., Gonzalez-Mendez, L. X., Valdes-Galicia, J. F., et al. (2024). The Mother's day geomagnetic storm on 10 may 2024: Aurora observations and low latitude space weather effects in Mexico. *Space Weather*, 22(11), e2024SW004111. Retrieved from <https://doi.org/10.1029/2024SW004111>
- Kappenman, J. G. (2007). *The electric power engineering handbook* (2nd ed.). CRC/IEEE Press. p. 16:1-22.
- Kwak, Y.-S., Kim, J.-H., Kim, S., Miyashita, Y., Yang, T., Park, S.-H., et al. (2024). Observational overview of the may 2024 g5-level geomagnetic storm: From solar eruptions to terrestrial consequences. *Journal of Astronomy and Space Sciences*, 41(3), 171–194. <https://doi.org/10.5140/JASS.2024.41.3.171>
- Lehtinen, M., & Pirjola, R. (1985). Currents produced in earthed conductor networks by geomagnetically-induced electric fields. *Annales Geophysicae*, 3(4), 479–484.
- McNish, A. G. (1940). The magnetic storm of march 24, 1940. *Terrestrial Magnetism and Atmospheric Electricity*, 45(3), 359–364. Retrieved from <https://doi.org/10.1029/TE045i003p00359>
- O'Callaghan, J., & Billings, L. (2024). The strongest solar storm in 20 Years did little damage, but worse space weather is coming. Retrieved from <https://www.scientificamerican.com/article/the-strongest-solar-storm-in-20-years-did-little-damage-but-worse-space/>
- Parker, W. E., & Linares, R. (2024). Satellite drag analysis during the may 2024 gannon geomagnetic storm. *Journal of Spacecraft and Rockets*, 61(5), 1412–1416. <https://doi.org/10.2514/1.A36164>
- Trivedi, N. B., Vitorello, I., Kabata, W., Dutra, S. L. G., Padilha, A. L., Bologna, M. S., et al. (2007). Geomagnetically induced currents in an electric power transmission system at low latitudes in Brazil: A case study. *Space Weather*, 5(4), n/a–n/a. Retrieved from <https://doi.org/10.1029/2006sw000282>